Entangled Photons for Violation of the CHSH Inequality

Rod Rofougaran
Physics Department, Columbia University
(Dated: October 30th, 2024)

ABSTRACT

This report presents an experimental investigation into the violation of a specific form of Bell's inequality, the Clauser-Horne-Shimony-Holt (CHSH) inequality, using entangled photons generated through spontaneous parametric down-conversion (SPDC). By measuring the polarization states of entangled photons across varying bases, quantum correlations were calculated to determine the CHSH parameter $|S|$. Multiple trials with varying integration times consistently yielded values of $|S| \geq 2$ with high statistical significance, demonstrating a clear violation of local hidden variable theories and confirming the non-locally real nature of entanglement. The highest obtained value of $|S| = 2.286 \pm 0.0068$, indicating non-local quantum behavior with 42 standard deviations above the classical bound.

I. INTRODUCTION

In their infamous 'EPR' paper published in 1935, Einstein, Podolsky, and Rosen suggested that the quantum mechanical description of physical reality may be incomplete [1]. They proposed a thought experiment involving a pair of entangled particles separated by a large distance. According to quantum mechanics, measuring an observable of one particle would instantaneously affect the observable of the other, which challenges local realism—a fundamental assumption in classical physics and Einstein's theory of relativity. Local realism suggests that physical properties exist independently of measurement and that information cannot travel faster than the speed of light. To address this paradox, later termed the 'EPR Paradox', the authors speculated that a theory of hidden variables might explain quantum mechanical phenomena.

In response to the EPR paper, John Stewart Bell, in 1964, while searching for a consistent hidden variable theory, theoretically proved that no hidden variable description of quantum mechanics is compatible with entanglement [2]. He achieved this by analyzing the quantum correlations of entangled particles, defined as the expectation value of the product of the measurements from both particles in an experiment. Bell demonstrated that these correlations are bounded if any hidden variable theory were to be correct, leading to the formulation of Bell's inequality. If measurements of entangled particles produce correlations that exceed this bound (known as a Bell Test), it would experimentally verify that quantum mechanics is inherently non-locally real.

This lab report investigates a violation of one form of Bell's inequality, specifically the CHSH inequality, as described by Clauser, Horne, Shimony, and Holt in their 1969 paper [3]. Since then, numerous Bell tests have been conducted to demonstrate violations of the CHSH inequality, most of which utilized entangled photons, including the experiments detailed in this report. A photon can be considered a two-level quantum system, where its polarization states represent these two levels. In other words, a photon can exist in a superposition of two distinct polarization states, such as horizontal and vertical, represented as:

$$|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \tag{1}$$

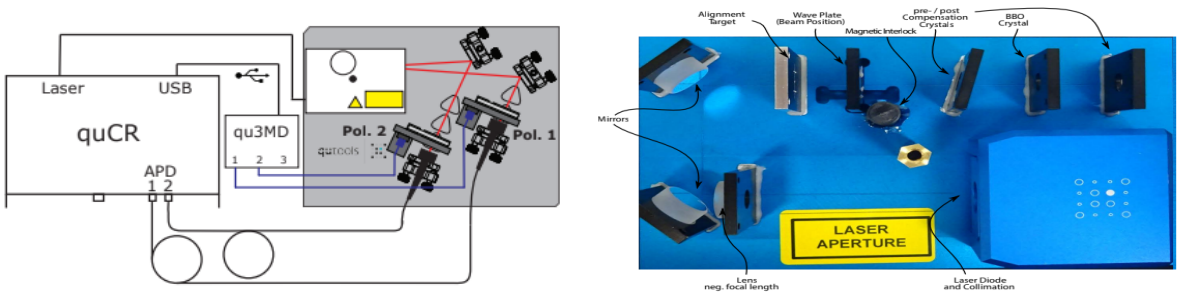
With this representation of our quantum system, multiple photons can become entangled with one another. By measuring the polarization of these entangled photons in different bases, we can calculate correlations that can be used to violate the CHSH inequality, which experimentally demonstrates the non-local nature of entanglement.

II. METHODS

A. Apparatus

The qutools Entanglement Demonstrator (quED) [4] is the apparatus used to perform this experiment, as shown in Figure 1a. The inner details of the laser source aperture, responsible for the preparation of entangled photons, are illustrated in Figure 1b. Initially, the laser diode emits ultraviolet photons that are linearly polarized in the vertical direction. The divergent light from the diode is focused using a collimating spherical lens and a negative spherical lens. The light then passes through a half-wave plate (the alignment target is used only for alignment and is removed during experimentation). The half wave plate rotates the polarization by 45° such that there are equal amounts of vertically and horizontally polarized photons. Thus, the polarization state of the photons can be described by the quantum state in Equation 1.

The beam of vertically and horizontally polarized photons encounters a pre-compensation birefringent crystal, non-linear down-conversion crystals (BBO), and a post-compensation birefringent crystal. The BBO crystals are responsible for a second-order nonlinear process known as spontaneous parametric down-conversion (SPDC). SPDC is a low-probability process that converts



(a) (b)

FIG. 1: Apparatus. 1a) Setup for measuring the CHSH inequality with the motorized quED. 2b) Inner details of the pump assembly with the nonlinear BBO crystal.

an intense laser beam into pairs of lower-frequency photons while conserving energy and angular momentum. This experiment utilizes two crystals that operate in a type-I phase-matching configuration, with each crystal oriented so that its optical axis lies in mutually perpendicular planes.

The consequence of type-I coupling is that the SPDC process occurs only in the crystal where the incoming photon is polarized along the crystal's optical axis, emitting two lower-frequency photons that are polarized perpendicular to that axis. Therefore, by aligning the optical axes of the two crystals to be horizontal and vertical, the type-I SPDC process results in the following transformation of incoming photons:

$$\text{SPDC}(|H\rangle) = |V\rangle_1 |V\rangle_2 , \text{SPDC}(|V\rangle) = |H\rangle_1 |H\rangle_2 \quad (2)$$

The pre- and post-compensation birefringent crystals in the quED are included to make sure the SPDC emission processes of the two crystals are mutually coherent. If the two processes are mutually coherent, we can apply the transformation in Equation 2 to the incoming quantum state $|+\rangle$ to produce the entangled state $|\Phi_+\rangle$

$$|\Phi_+\rangle = \text{SPDC}(\frac{1}{\sqrt{2}}(|H\rangle+|V\rangle)) = \frac{1}{\sqrt{2}}(|V\rangle_1 |V\rangle_2+|H\rangle_1 |H\rangle_2) \quad (3)$$

Since the SPDC process occurs with low probability, high-frequency light is filtered out using a long-pass filter as the photons exit the laser aperture. This ensures that only the low-frequency down-converted photons are collected. As shown in Figure 1a, once the lower-frequency entangled photons exit the long-pass filter, each photon of the entangled pair is reflected off a mirror and encounters polarizers 1 and 2. Each polarizer is motorized and rotatable via the qu3MD unit. The measured photons are then collected by the fiber couplers connected to the quCR unit for the calculation of correlations using coincidence counts.

B. Experimental Procedure

In conducting this experiment, the majority of the time was spent on alignment. The goal of the alignment process was to maximize the number of entangled photons that the couplers could simultaneously detect, while minimizing noise. The components of the apparatus being aligned include two mirrors and two fiber couplers, each with two knobs corresponding to vertical and horizontal control, resulting in a total of eight degrees of freedom (DOF).

Two separate alignment procedures were conducted: rough alignment and fine alignment. Rough alignment was performed first, as described below and illustrated in Figure 2. For this procedure, the white cover over the laser source aperture was removed, the half-wave plate was taken out, and the single-mode fibers were disconnected from the quCR unit. Visible red alignment lasers were then connected to the single-mode fibers. The laser source was turned on, and the alignment plate was inserted so that the blue pump laser intersected the center of the alignment plate. The alignment lasers were then activated, and each mirror was aligned one at a time to ensure that the two red beams intersected the left and right engraved alignment crosses.

Once the rough alignment was deemed satisfactory, the white cover was replaced, the alignment helper was removed, and the fibers were reconnected to the quCR unit to proceed with fine alignment. During this process, the quality of alignment was assessed based on single count rates and coincidence count rates measured by the quCR. Initially, single count rates were maximized by adjusting the knobs on the couplers, each providing two DOF. Next, for each coupler-mirror pair, an iterative angular alignment procedure, commonly known as 'walking', was performed to maximize the coincidence rates. During this stage, entanglement experiments were conducted using the quCR, and based on the quality of the entanglement, either fine or rough alignment was repeated as necessary. The complete alignment process took approximately three weeks.

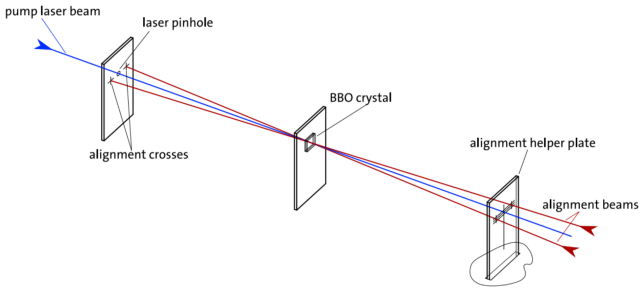


FIG. 2: Visualization of satisfactory rough alignment using the pump and alignment beams and alignment helpers.

C. CHSH Inequality Theory

The CHSH inequality, the violation of which can only be explained by the incompatibility of entanglement with local hidden variable theories, is expressed as follows:

$$|S| \leq 2, \tag{4}$$

where

$$S(\alpha, \alpha', \beta, \beta')| = E(\alpha, \beta) + E(\alpha', \beta) - E(\alpha, \beta') + E(\alpha', \beta') \tag{5}$$

and $(\alpha, \alpha', \beta, \beta')$ denote the local measurement settings of the two observers. Measurement settings in our experiment correspond to the polarization angles of polarizers 1 and 2, so $(\alpha, \alpha', \beta, \beta')$ denote these angles. Quantum mechanical theory predicts that the maximum violation of the CHSH inequality occurs when $(\beta - \alpha) = (\alpha' - \beta) = (\beta' - \alpha') = 22.5°$ and $(\beta' - \alpha) = 67.5°$, where $|S|_{max} = 2\sqrt{2} \geq 2$. $|S|_{max}$ is achieved when our entangled state is maximally entangled. We will use these angles to verify this violation experimentally.

Let $E(a, b)$ be the expectation value of $A(a) * B(b)$ where A, B are measurement outcomes with measurement settings a, b respectively. If the measurement outcomes for each observer are ± 1 for measuring state $|\pm\rangle$, then expectation value of the product of the measurement outcomes is given by:

$$E(a, b) = P_{++}(a, b) - P_{+-}(a, b) - P_{-+}(a, b) + P_{++}(a, b) \tag{6}$$

In our experiment, in order to determine these probabilities, the quCR unit calculates coincidence counts, denoted as $C(a, b)$. These are the counts of simultaneous detection of the photons by the two fiber coupler detectors, when the polarizers are set to angles a and b. To detect measurement outcomes (-1), we rotate our angle a or b by 90° denoted by a_- and b_-. Therefore, our normalized probabilities in Equation 6 are calculated using coincidence counts by the following equation:

$$P_{\pm\pm} = \frac{C(a_{\pm\pm}, a_{\pm\pm})}{\text{Total Counts}(a, b)} \tag{7}$$

where $a_+ = a$ corresponding to measurement outcomes of $(+1)$. Plugging in our probabilities into Equation 6, we can solve for $E(a, b)$ in terms of coincidence counts given by:

$$E(a, b) = \frac{C(a_+, b_+) - C(a_+, b_-) - C(a_-, b_+) + C(a_-, b_-)}{C(a_+, b_+) + C(a_+, b_-) + C(a_-, b_+) + C(a_-, b_-)} \tag{8}$$

As we can see, for each (a, b), we need counts from four separate sets of polarizer angle settings, which results in a total of 16 measurements settings for $|S|$ calculation.

The errors $dE(a, b)$ can be calculated using Gaussian error propagation to obtain the following equation:

$$dE(a, b) = 2\frac{[C(a_+, b_+) + C(a_-, b_-)][C(a_+, b_-) + C(a_-, b_+)]}{(C(a_+, b_+) + C(a_+, b_-) + C(a_-, b) + C(a_-, b_-))^2} \times \left(\frac{1}{C(a_+, b_+) + C(a_-, b_-)} + \frac{1}{C(a_+, b_-) + C(a_-, b_+)}\right) \tag{9}$$

Finally, using error propagation of sums, we can determine the error associated with our experimental value of S to be:

$$dS(\alpha, \alpha', \beta, \beta') = \sqrt{\sum_{a=\alpha,\alpha'} \sum_{b=\beta,\beta'} (dE(a, b))^2} \tag{10}$$

D. Corelation Curves Theory

In order to measure $|S|$, we only need the correlations of discrete measurement angles. Another way of verifying the entanglement of pairs of photons is by measuring correlations with one of the measurement angles varying continuously from 0 to 2π, while keeping the other fixed. Let us assume we keep α fixed and vary β, while preparing the state $|\Phi_+\rangle$. If we measure the first photon with respect to linear polarization $|\alpha\rangle = \frac{1}{\sqrt{2}}(\cos\alpha\,|H\rangle + \sin\alpha\,|V\rangle)$ and the second photon with respect to linear polarization $|\beta\rangle = \frac{1}{\sqrt{2}}(\cos\beta\,|H\rangle + \sin\beta\,|V\rangle)$, we can calculate the probability of both photons passing through the polarizers as a function of β

$$\begin{aligned} P_{\alpha,|\Phi_+\rangle}(\beta) &= |\langle\alpha| \otimes \langle\beta|\,|\Phi_+\rangle|^2 \\ &= \frac{1}{2}|\cos\alpha\cos\beta + \sin\alpha\sin\beta|^2 \\ &= \frac{1}{2}\cos^2(\alpha - \beta) \end{aligned} \tag{11}$$

Therefore, we expect to measure correlation curves that exhibit the shape of $\cos^2$, maximized at $\alpha = \beta$ for $|\Phi_+\rangle$. In contrast, separable states yield different correlation curves. For instance, consider the state $|HH\rangle$. If we calculate $P_{\alpha,|HH\rangle}$ in a manner similar to that of Equation 11, we derive the following equation:

$$P_{\alpha,|HH\rangle}(\beta) = |\langle\alpha| \otimes \langle\beta| |HH\rangle|^2 = \frac{1}{2}\cos^2(\alpha)\cos^2(\beta) \tag{12}$$

From this equation, we observe that the amplitude of oscillation of $P_{\alpha,|HH\rangle}$ depends on the value of α. For example, if we fix α at 90°, we should expect to see no oscillation for the separable state $|HH\rangle$. In contrast, for a maximally entangled state, such as $|\Phi_+\rangle$, the amplitude of oscillation is maximized at the same value, where $\alpha = \beta$, regardless of the choice of α. This distinction allows us to qualitatively measure the degree of entanglement of a state by observing how the amplitudes of oscillation vary with respect to our fixed measurement angle α.

III. RESULTS AND DISCUSSION

A. CHSH Inequality Experiment

We conducted multiple experiments to calculate $|S|$ with varying integration times ranging from 1 s to 10 s, and the results are shown in Figure 3. The integration time corresponds to the duration for which each of the 16 measurement settings allows the motorized polarizer to accept incoming photons. Consequently, longer integration times result in higher coincidence counts. As illustrated in Figure 3, increased integration times also lead to tighter error bounds, which is expected, as a higher number of coincidence counts generally results in more accurate measurements. Although no dependence was observed between integration times and the value of $|S|$, all experiments consistently yielded $|S| \geq 2$.

To illustrate the process of calculating $|S|$, we present sample calculations of correlations E and dE in Table I from the experiment with the lowest dS at an integration time of 10 s. The coincidence counts recorded for each measurement setting were output directly from the quCR unit as raw data, and the values of E and dE were calculated using Equations 8 and 9. By substituting these values into Equation 10, we can determine our experimentally obtained $|S|$ value for this experiment to be:

$$|S| = 2.286 \pm 0.0068 \tag{13}$$

This experiment demonstrated non-local entanglement with high certainty, as our $|S|$ value was approximately 42 standard deviations above the hidden variable bound of $|S| = 2$. However, we were unable to achieve maximal entanglement of $|S| = 2\sqrt{2}$. This limitation could stem from several factors.

The primary source of error was slight misalignment of the laser pump, mirrors, and fiber couplers. According to the inspection sheet for the quCR unit, the ideal 'singles' count rate of photons entering each polarizer should exceed 50,000 photons per second, with coincidence counts surpassing 3,000 photons per second. After following the rough and fine alignment procedures over multiple weeks, we achieved our best singles counts and coincidence counts of approximately 30,000 and 600 photons per second, respectively. Although these values were lower than ideal, they were still sufficient to violate the CHSH inequality with statistical significance. Another source of error in our experiment was environmental background noise. External light collected by the fiber couplers could increase the single and coincidence count rates, leading to inaccurate results. This background light originated from both within and outside the laboratory. To mitigate this, we conducted all trials with the laboratory lights turned off and attempted to block external light as much as possible.

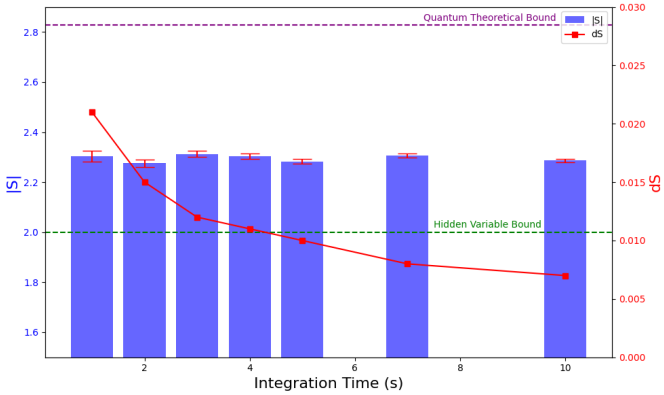


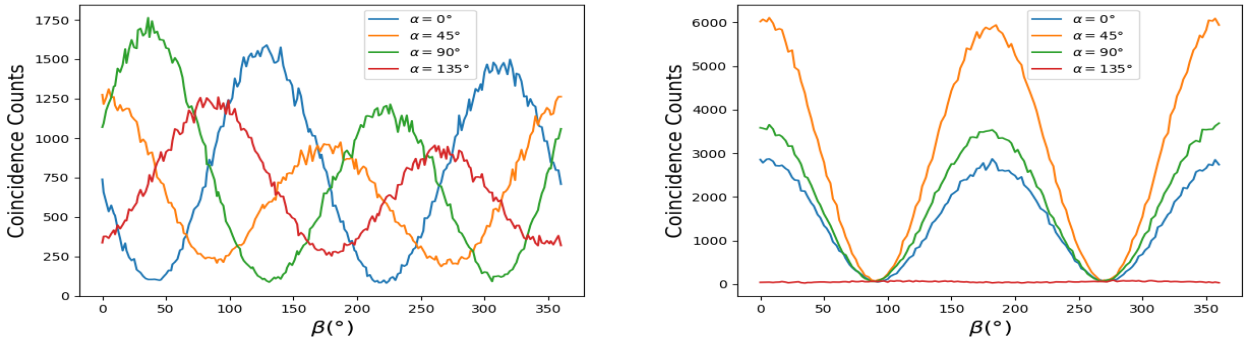
FIG. 3: $|S|$ values plotted against integration times. For each integration time, $S \geq 2$ with statistical significance. dS is also plotted against integration (red line), indicating a negative correlation.

B. Corelation Curves Experiment

Once we achieved high entanglement quality in our CHSH inequality experiments, we measured correlation curves for both entangled and non-entangled states. The non-entangled state was prepared by simply removing the half-wave plate. Using an integration time of 2 seconds, the quCR/qu3MD automatically conducted our correlation curve experiments by fixing the polarizer angle α and continuously rotating β from 0 to 2π. Four curves were measured for α fixed at 0°, 45°, 90°, and 135°, with the results shown in Figure 4. As seen in Figure 4(a), each of the four correlation curves exhibits a $\cos^2$ shape. The amplitudes of the four curves differ reasonably, indicating that the entangled state is not maximally entangled. More precisely, not all of the coincidences recorded were from maximally entangled photon pairs. Despite this, we can clearly distinguish our entangled state correlation curves from our benchmark non-entangled state correlation curves in Figure 4(b). For our separable state, the

	Polarizer 1 (°)	Polarizer 2 (°)	C (photons/s)	E	dE
(α, β)	0	22.5	3682	-0.7816	0.00244
	0.0	112.5	28132		
	90.0	22.5	29947		
	90.0	112.5	3437		
(α', β)	45.0	22.5	21354	0.4247	0.00376
	45.0	112.5	6606		
	135.0	22.5	10059		
	135.0	112.5	19918		
(α, β')	0.0	67.5	7080	-0.5856	0.00333
	0.0	157.5	23020		
	90.0	67.5	23961		
	90.0	157.5	5200		
(α', β')	45.0	67.5	6143	-0.4938	0.00387
	45.0	157.5	16477		
	135.0	67.5	21247		
	135.0	157.5	6641		

TABLE I: Measured coincidence counts C, calculated correlations E, and calculated errors dE, for polarizer angles α, α', β, $\beta' = 0°, 45°, 22.5°, 67.5°$.



(a) (b)

FIG. 4: Correlation curves for the a) entangled and b) separable state prepared and collected by the quED. β denotes the continuously varying polarizer 2 angle, while the polarizer 1 angle α is fixed at 0°, 45°, 90°, and 135°

amplitude of the curves fully depend on α, indicating the state is indeed separable.

However, an interesting unexpected result arises from these two sets of correlation curves, particularly evident in our non-entangled state. Assuming the laser pump emits vertically polarized light and based on our knowledge of type-I SPDC from the BBO crystal, we would expect that in the absence of the half-wave plate, $|V\rangle$ should be down-converted to $|HH\rangle$. If this were the case, using Equation 12, we would anticipate the maximum amplitude curve at $\alpha = 0°$ and an all-zero curve at $\alpha = 90°$. Instead, our maximum amplitude curve and all-zero curve correspond to $\alpha = 45°$ and $\alpha = 135°$, respectively. This indicates that our collected state is, in fact, the separable state $|{+}{+}\rangle$. Furthermore, following from Equation 11, we expect the correlation curves of the entangled state to be maximized when $\beta = \alpha$, but we observe that the curves are maximized when $\beta = \alpha - 45°$. This suggests that our entangled state is approximately $\frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$, not $|\Phi_+\rangle$. These unexpected results could be explained by the SPDC process down-converting in the diagonal basis rather than the standard basis, but further experimentation is required to verify this. Nevertheless, our successful result in violating the CHSH inequality and distinguishing between entangled and non-entangled states using correlation curves is independent of the basis we are working with.

IV. CONCLUSION

In conclusion, using the qutools Entanglement Demonstrater apparatus, we successfully experimentally violated the CHSH Inequality with statistical significance using entangled photons. This indicates that the entanglement of these photons cannot be explained by a hidden variable variable, implying that quantum mechanics is inherently non-locally real.

[1] A. Einstein, B. Podolsky, and N. Rosen, Can quantum-mechanical description of physical reality be considered complete?, Phys. Rev. **47**, 777 (1935).

[2] J. S. Bell, On the einstein podolsky rosen paradox, Physics Physique Fizika **1**, 195 (1964).

[3] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed Experiment to Test Local Hidden-Variable Theories, Phys. Rev. Lett. **23**, 880 (1969).

[4] quTools, *quED Entanglement Demonstrater Manual*, quTools.com.